GeoFly3D: Terrain Visualization with Autonomous Drones

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Parrot Anafi 4K

Introduction

- This project focuses on the autonomous scanning and mapping of terrestrial land features using drones, producing high-quality 3D schemes and 2D orthophotographic models.
- The primary objective is to develop an efficient photogrammetric pipeline that enhances the quality of the resulting 2D and 3D models. Additionally, the project seeks to integrate a multi-drone feature into a single mission, significantly reducing mission time while maintaining robust modeling performance.
- Aimed at geologists, this product will enable detailed analysis of post-earthquake landscapes and facilitate the scanning of large territories with improved quality and cost-effectiveness.

Area Division & Route Creation

In order to decrease the mission's duration, we focused on improving the flight itinerary's optimality and the inclusion of multiple drones in a mission. This process can be explained by dividing it into two topics:

Route Creation: Many papers recommended parallelly aligning the route with the longest edge on the selected shape's periphery. In addition, the importance of adding a complementary route which is rotated with respect to the optimal route was underscored as a means of improving the photogrammetric models' robustness. Hence, both routes were implemented to encompass the whole area. The theoretical premise rests on intersecting multiple parallel lines with the perimeter of the shape. The lines are separated by the following distance:

$$displacement = 2 * (1 - intersection\ ratio) * flight\ height * \tan\frac{HFOV}{2}$$

Area Division: Although the simulator was unable to support multi-drone test settings, it allows users to counterfactually observe the division of labor among multiple drones. The optimization function ensures that the subregions can be collectively scanned in minimal time. To achieve this, the implementation requires estimating route time and using a heuristic for area selection. To divide the area into 'N' subregions, let's denote the subregions as:

$$A_1, A_2 \ldots A_N$$

and the lines that separate the subregions and their center coordinates as:

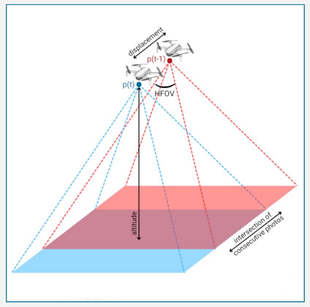
$$L_1, L_2, \ldots\ldots, L_{N-1} \qquad Center(L_i) = (X_i, Y_i)$$

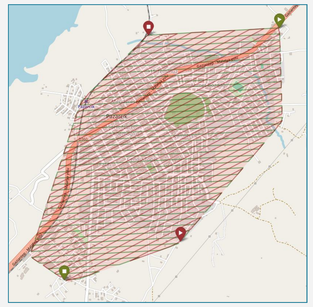
The division operation is performed through several iterations. During each iteration, each line object checks the adjacent region and compares the estimated route time. If one region has a higher time estimation, the line moves in that direction until the time estimations are equal. Here, 'k' is the coefficient for the direction of movement, 'd' is the length of the diagonal of the bounding box of the total area, and 'M' is the slope of the diagonal.

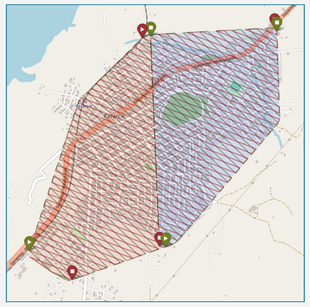
$$X_i(t+1) = X_i(t) + (k_{dir} * ratio * d_{diagonal} \cos(\tan^{-1}(M_{diagonal})))$$
$$Y_i(t+1) = Y_i(t) + (k_{dir} * ratio * d_{diagonal} \sin(\tan^{-1}(M_{diagonal}))) \quad \text{if} \quad T_{est}(A_i) \not\equiv T_{est}(A_{i+1})$$

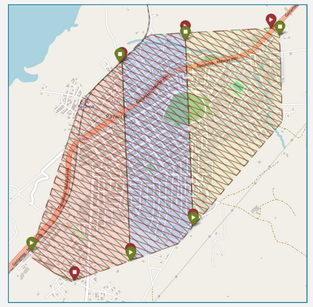
The formula for route time estimation is shown below.

$$T_{est} = \frac{Distance}{Speed_{avg}} + (NodeCount * 4)$$









Results

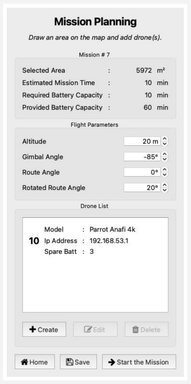
For the purpose of assessment, reference objects have been utilized. The three primary references are the width of the field, a soccer goal post, and a carton board.

	Ground Truth	Result	Error
Soccer Goal	750 cm	725 cm	-25 cm
Box Width	49 cm	43 cm	-6 cm
Box Height	53 cm	66 cm	13 cm
Wield Width	71 m	71.11 m	11 cm

Although the error rates in the images varied, the overall error remained submeter regardless of the size of the object. Consequently, providing a percentage error rate is challenging, as it has been observed that the error does not increase proportionally with the size of the object.

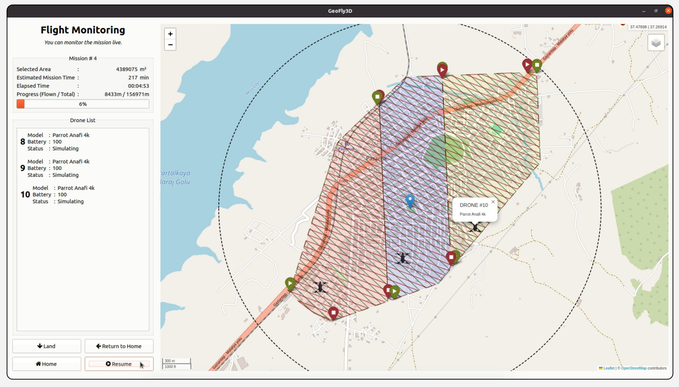
Furthermore, the lack of a proportional increase in error with object size suggests that this method is particularly well-suited for diverse environments and objects of varying dimensions. This characteristic is especially beneficial for applications requiring high precision and detailed analysis, such as geological surveys and post-disaster assessments.

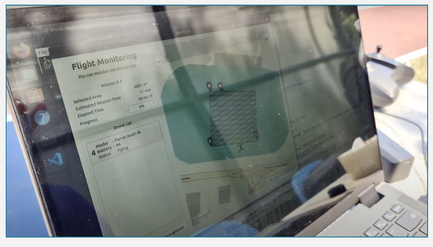
Mission Planning & Real-Time Tracking



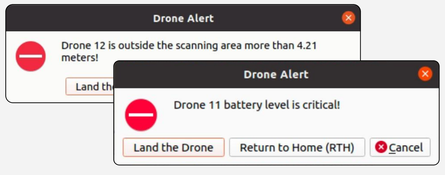
Automating and augmenting the efficiency of mission planning and drone tracking in landscape survey expeditions have been the core goals of our project. Those targets were tackled with a user-friendly application as follows:

- **Mission Planning**: The desktop application allows the user to create a mission by selecting the area to be scanned, defining the drone models involved in the mission, recording the number of spare batteries, and determining the shooting angle. Upon mission creation, the user can observe the routes created on their screen.
- **Mission Tracking:** During the mission time, the system live streams real-time data and provides the users with continuous updates regarding the battery's status, drone's position, drone's status and the mission's progress.





- **Operational Flexibility:** Due the unforeseeable circumstances, the field researchers are usually obliged to temporarily pause the mission. Nevertheless, the application stores mission information and enables the users to resume the mission afterwards.
- **Emergency Handling:** In addition to providing emergency alerts in case of a contingency, the application provides the user with options so that they do not lose control.

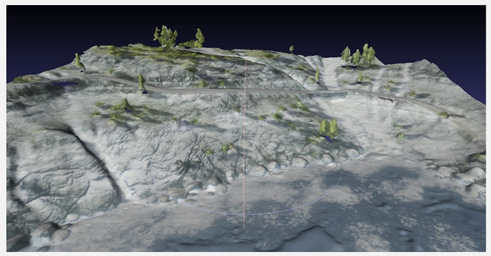


2D & 3D Model Generation

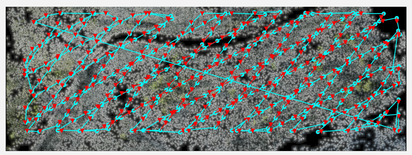
In the realm of terrestrial scanning, many models and techniques are employed to better visualize the landscape. This project focused on two of the most ubiquitously utilized models: 2D orthophotographic models and 3D models

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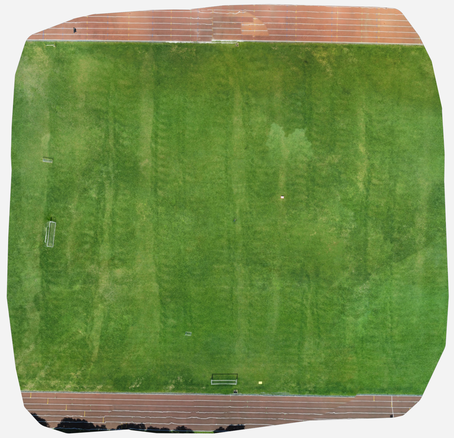
- **2D orthophotographic models**: They provide distortion-free 2D models of the terrain. One of their most important application is surveying and modeling natural disaster-stricken landscapes. Moreover, they can be used in urban planning and infrastructure planning.
- **3D models**: In addition to the 2D models, 3D models provide a clearer view of the terrestrial and topological changes in the regions afflicted by earthquakes. Moreover, they can be used to assess the severity of the damages incurred by building structures.



3D Model



Flight Path



Orthophoto of Uçaksavar Field